University of Minho

Academic Year: 2025/26

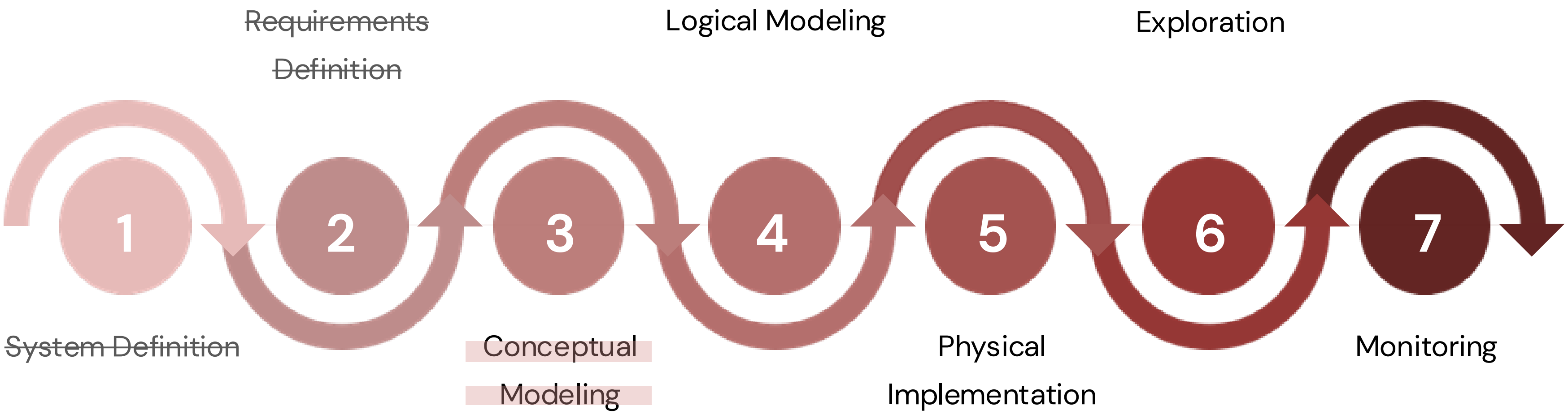
Databases

Conceptual Modeling (continued)

António Abelha | abelha@di.uminho.pt

01

Database Development Lifecycle



02

Database Development Lifecycle: Conceptual Modeling

Phase 1

IDENTIFY ENTITIES

Phase 2

IDENTIFY RELATIONSHIPS

Phase 3

IDENTIFY AND ASSOCIATE ATTRIBUTES

Phase 4

DEFINE ATTRIBUTE DOMAINS

Phase 5

IDENTIFY PRIMARY KEY ATTRIBUTES

(..)

03

Database Development Lifecycle: Conceptual Modeling

(..)

Phase 6

CONSIDER THE USE OF ENHANCED MODELING CONCEPTS (optional)

Phase 7

CHECK FOR REDUNDANCY

Phase 8

VALIDATE USER TRANSACTIONS

Phase 9

REVIEW AND VALIDATE THE CONCEPTUAL MODEL

PHASE 3: Conceptual Modeling

Enhanced Modeling Concepts

Specification/Generalization

- The concept of specialization/generalization is associated with the hierarchy of entities and special types of entities known as superclasses and subclasses, as well as the process of inheriting attributes.

- Specialized classes are called subclasses, while a generalized class is called superclass.

- A subclass inherits the stats of its superclass.

- A subclass is best understood through an "IS-A" analysis. Namely: "Engineer IS-A Worker", and "Card IS-A Payment Method".

- There are two aspects to analyze in the subclass/superclass relationship:
  - PARTICIPATION: full/mandatory or partial/optional
  - TYPE: disjunction (d) – {Or} or overlapping(o) – {And}

PHASE 3: Conceptual Modeling

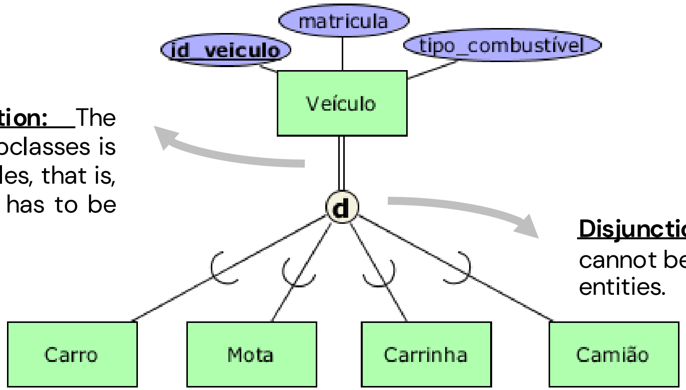
Enhanced Modeling Concepts

Specification/Generalization

Example: entities that are members of the "Vehicle" entity can be classified as "Car", "Motorbike", "Van" and "Truck". That is, the "Vehicle" entity is the superclass of the remaining entities.

- "Motorbike is a specification of "Vehicle". (increases granularity – top-down approach)
- "Vehicle" is a generalization of " Motorbike"; (decreases granularity – bottom-up approach)

Total/mandatory participation: The union of the entities of the subclasses is equal to the total set of vehicles, that is, the entire vehicle necessarily has to be one of the subclasses.



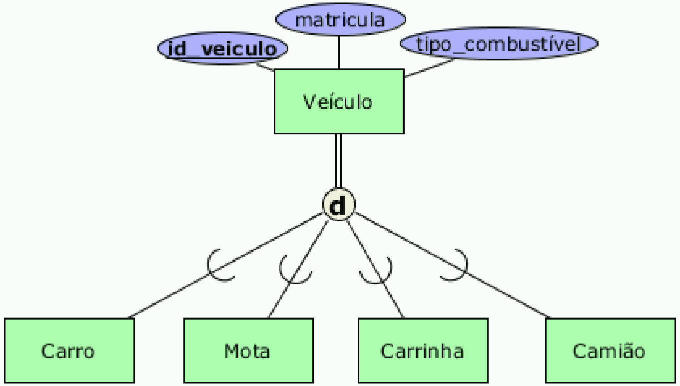
Disjunction: A vehicle is either a car or a motorcycle, it cannot be both simultaneously → there is no overlap of entities.

PHASE 3: Conceptual Modelling

➜ Enhanced Modeling Concepts

Specification/Generalization

TerraER



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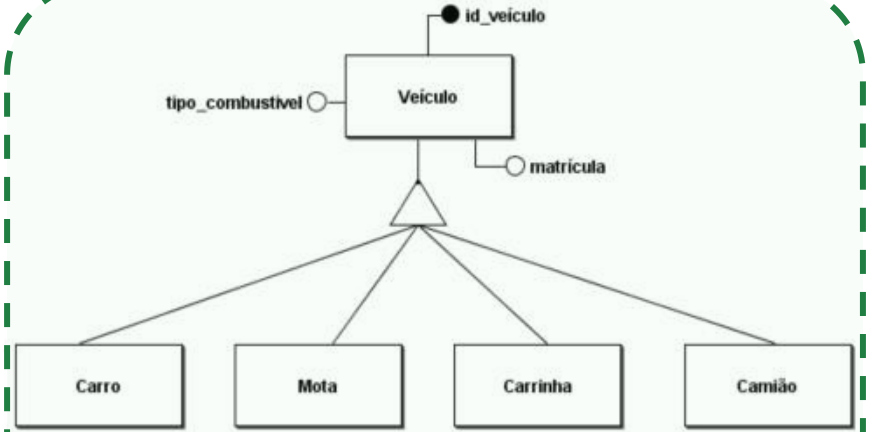


FASE 3: Modelação Conceptual

Case Study

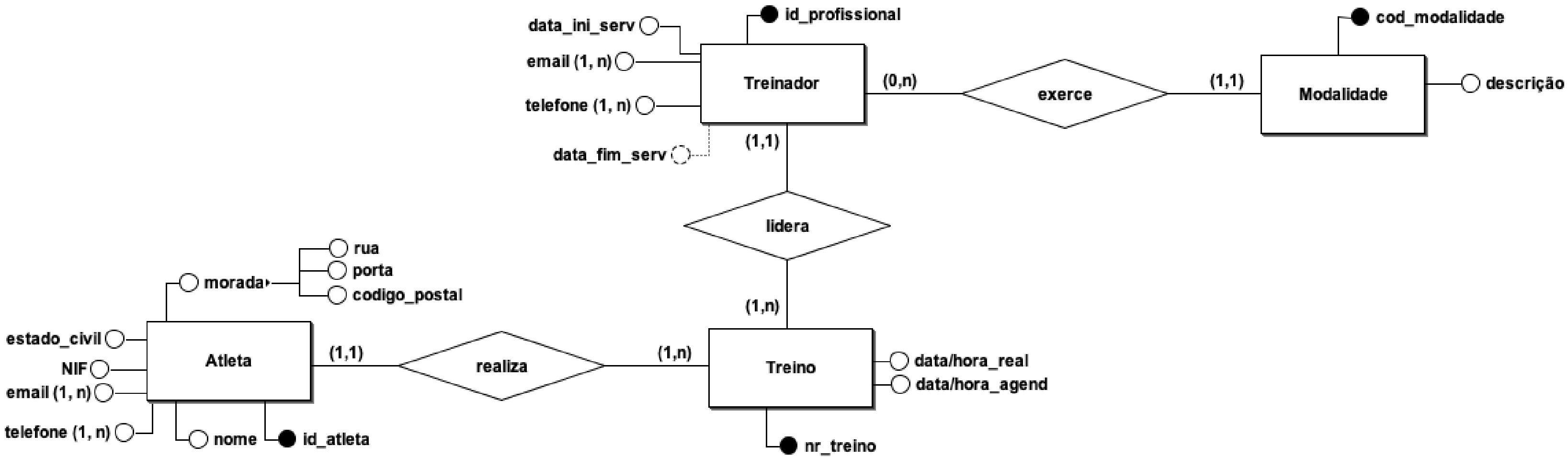
Question 6 : Build an E-R Diagram, using Chen's Notation, capable of meeting the following requirements:

- Each athlete at the center must be registered with the 6-digit athlete ID (unique value).
- It is necessary to store demographic data of the athlete, namely name, date of birth, address, telephone, email, tax number (NIF), ID, marital status, etc.
- An athlete can be coached by several coaches.
- A coach must be registered with his professional number, name, start date of service, end date, telephone contact and email address.
- A coach exercises one and only one modality.
- Each modality of the centre must be characterised by the modality code and its description.
- The training log must include the training number (unique identifier), the date and time of the schedule, the start date and end date of the training, the associated modality, etc.

PHASE 3: Conceptual Modelling

ER Diagram

Vista Treino A



PHASE 3: Conceptual Modelling

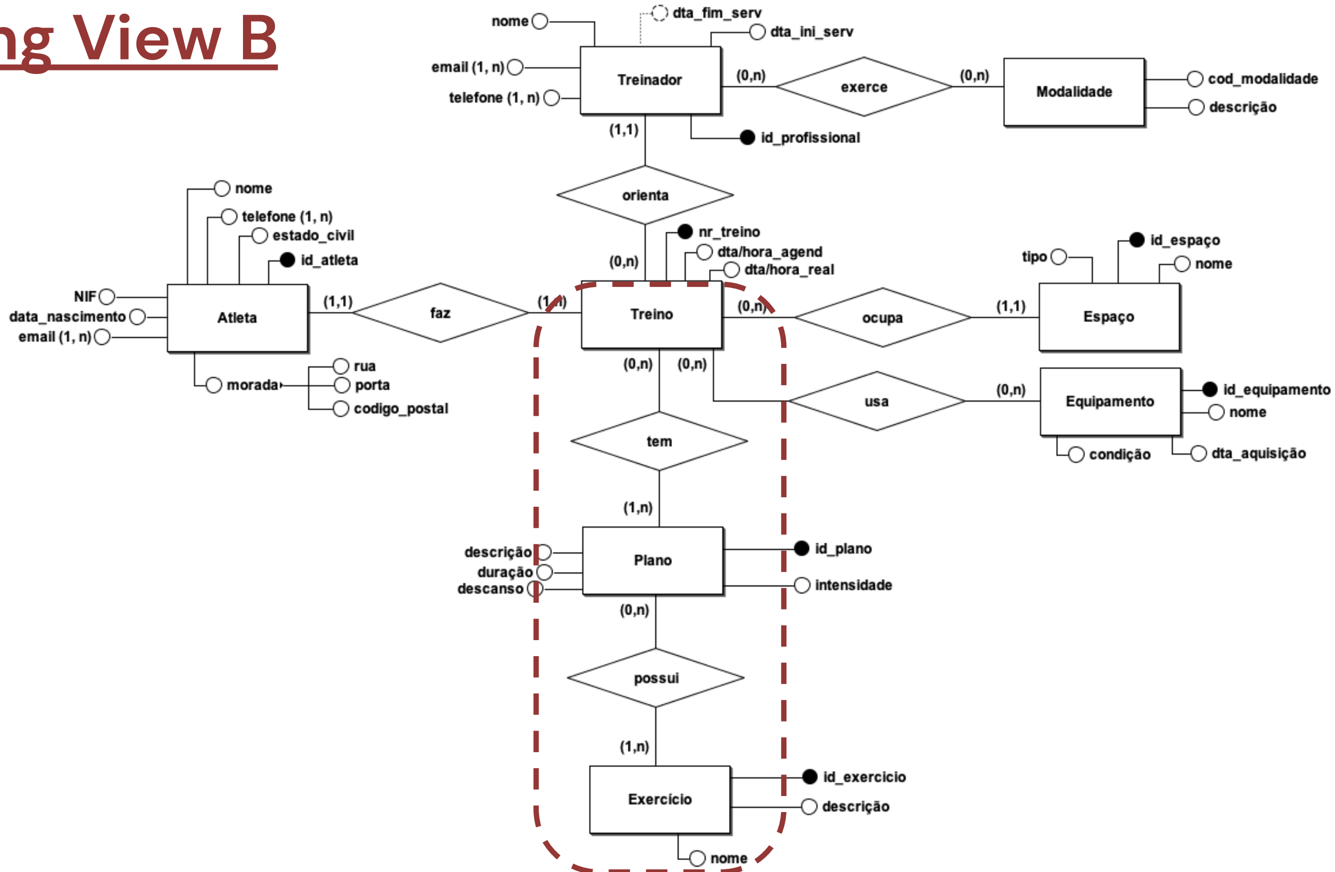
ER Diagram

Question 7: Modify the previous E-R Diagram to accommodate the following requirements:

- During a workout, certain equipment can be used.
- A piece of equipment must be registered with an identifier, the name and the current condition (Maintenance, Inactive, Damaged, Active...).
- Equipment is not necessarily disposable and, for this reason, there is equipment that can be used in more than one workout.
- A training session can be carried out in a specific space of the training center.
- The training center spaces must be registered with an identifier, the name and the type.
- The workout may have an associated plan.
- A plan can count for more than one exercise.
- The plan must contain the description, intensity, difficulty, duration and rest time.
- An exercise should be stored with an identifier, name, and description.

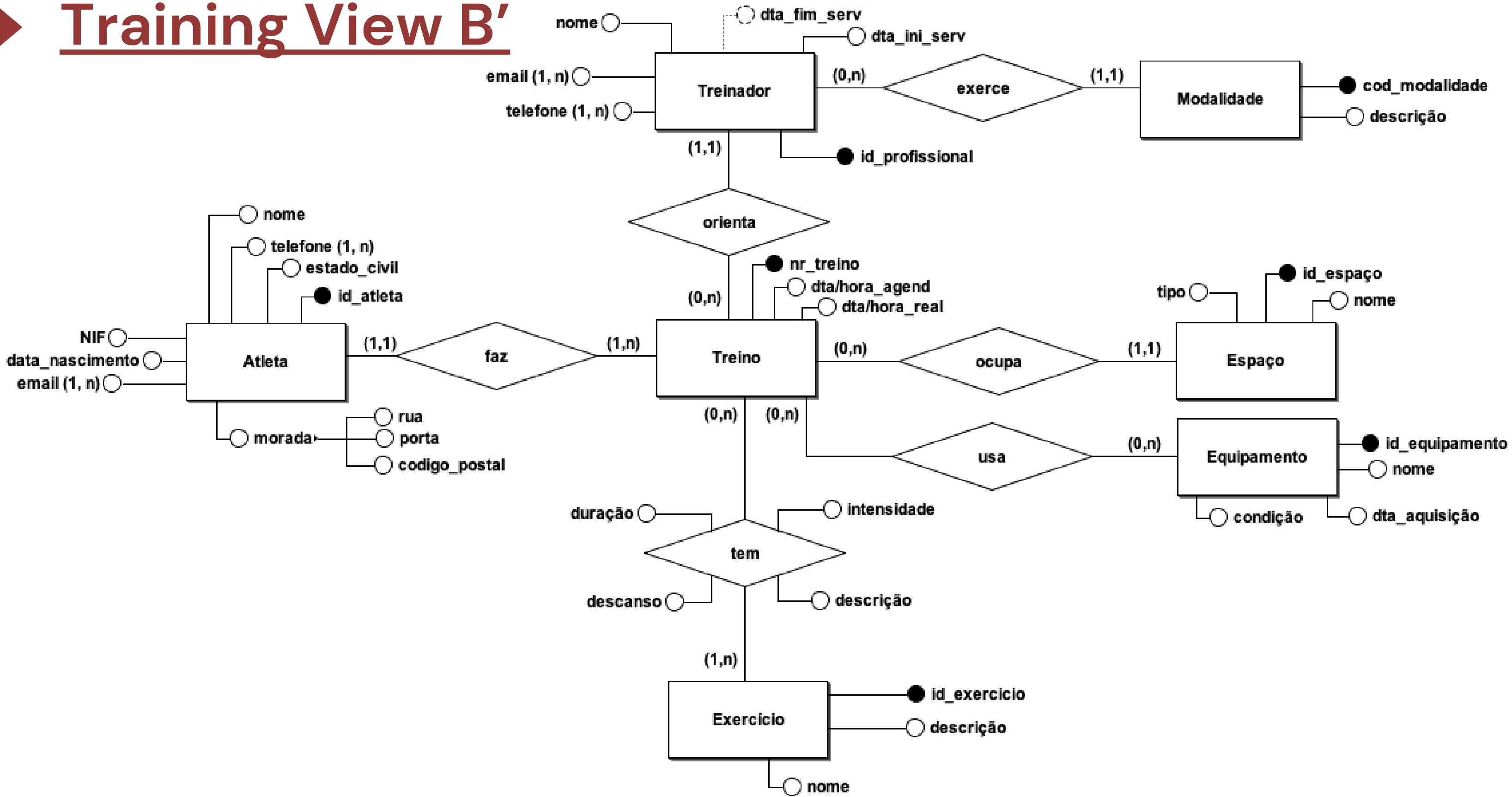
PHASE 3: Conceptual Modelling

Training View B



PHASE 3: Conceptual Modelling

➜ Training View B'



Usage Views

According to the requirements defined for the training center case study, the following usage views were identified:

USER FUNCTIONS

- Athletes;
- Coaches;
- Health Professionals;

$\left.\begin{array}{l}\text{Coaches;}\\ \text{Health Professionals;}\end{array}\right\}$ Staff

WORK AREA

- Consultations;
- Training Sessions; ✓

PHASE 3: Conceptual Modeling

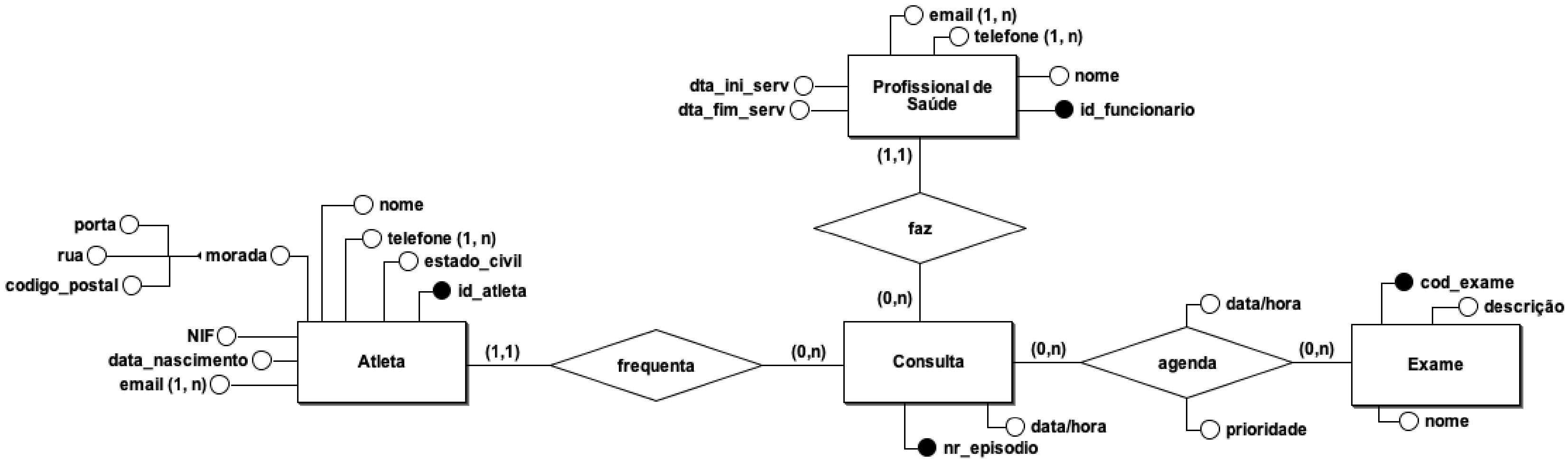
Caso de Estudo

Question 8: Build an E-R Diagram, using Chen's Notation, capable of meeting the following requirements:

- The appointment log should include the episode number (unique identifier), the athlete's identifier, date and time of the appointment, start date and end date of the appointment, the associated specialty, location, etc.
- At the time of consultation, professionals can schedule exams.
- Healthcare professionals should have the ability to record notes and detailed observations during consultations.
- The exam schedule must contain the date of the exam and priority (emergent, urgent, discharged, normal)
- An exam must be registered with the code, name, and type.
- Each healthcare professional must be registered with their employee ID (unique value), name, email, telephone, specialty, service start and end date.

PHASE 3: Conceptual Modeling

➜ Consultation view



PHASE 3: Conceptual Modeling

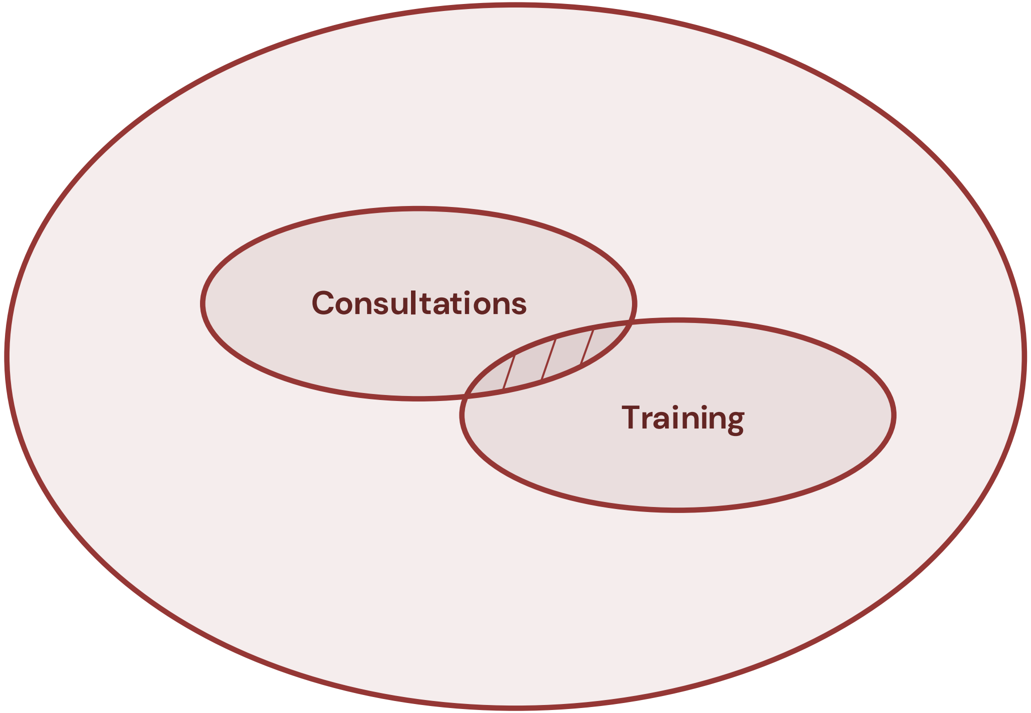
Case Study

Question 9: Combine the views in a single E-R model.

PHASE 3: Conceptual Modeling

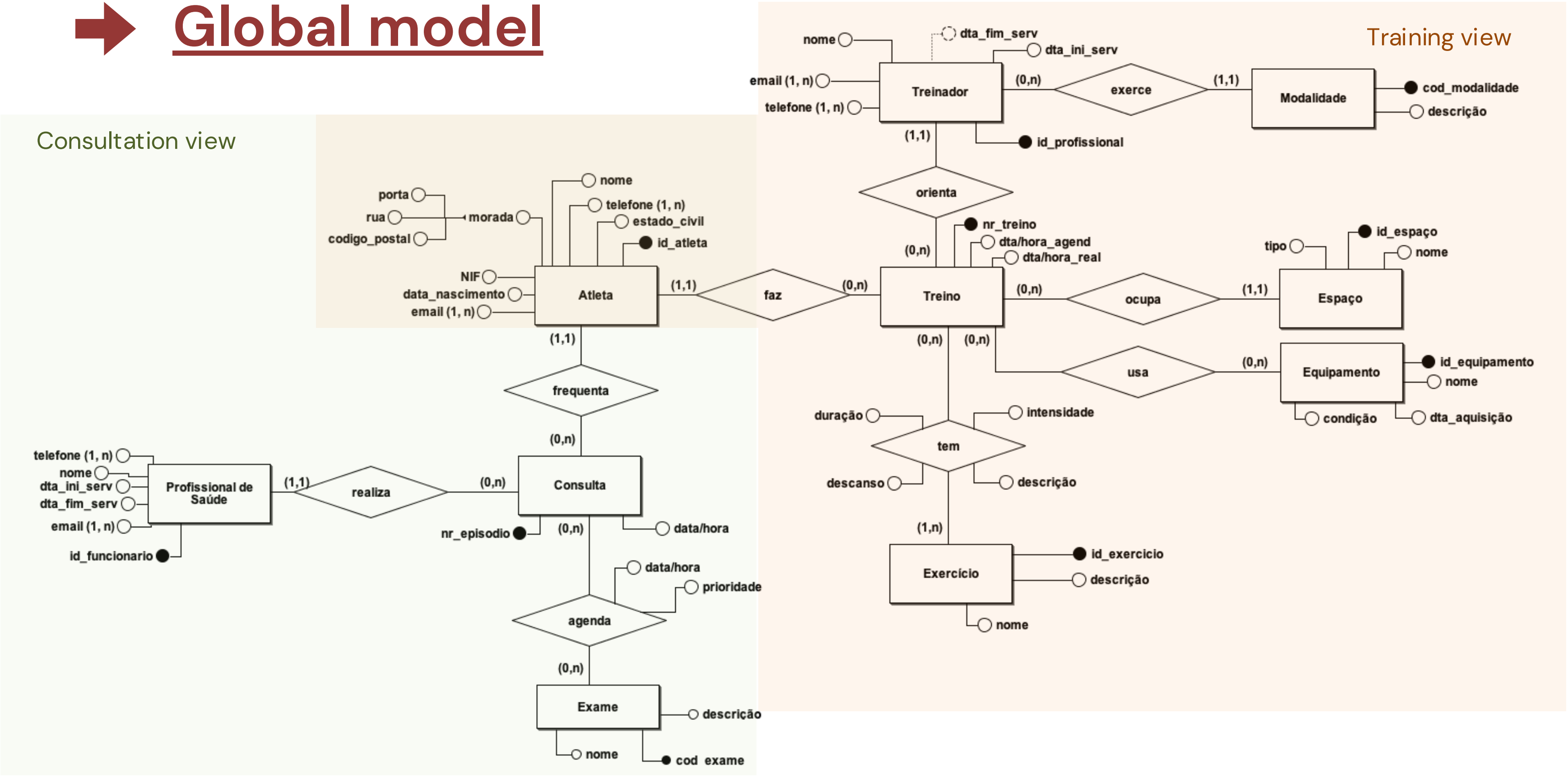
Final ER Model

(Combining views in a single ER model)



PHASE 3: Conceptual Modeling

➔ Global model



PHASE 3: Conceptual Modeling

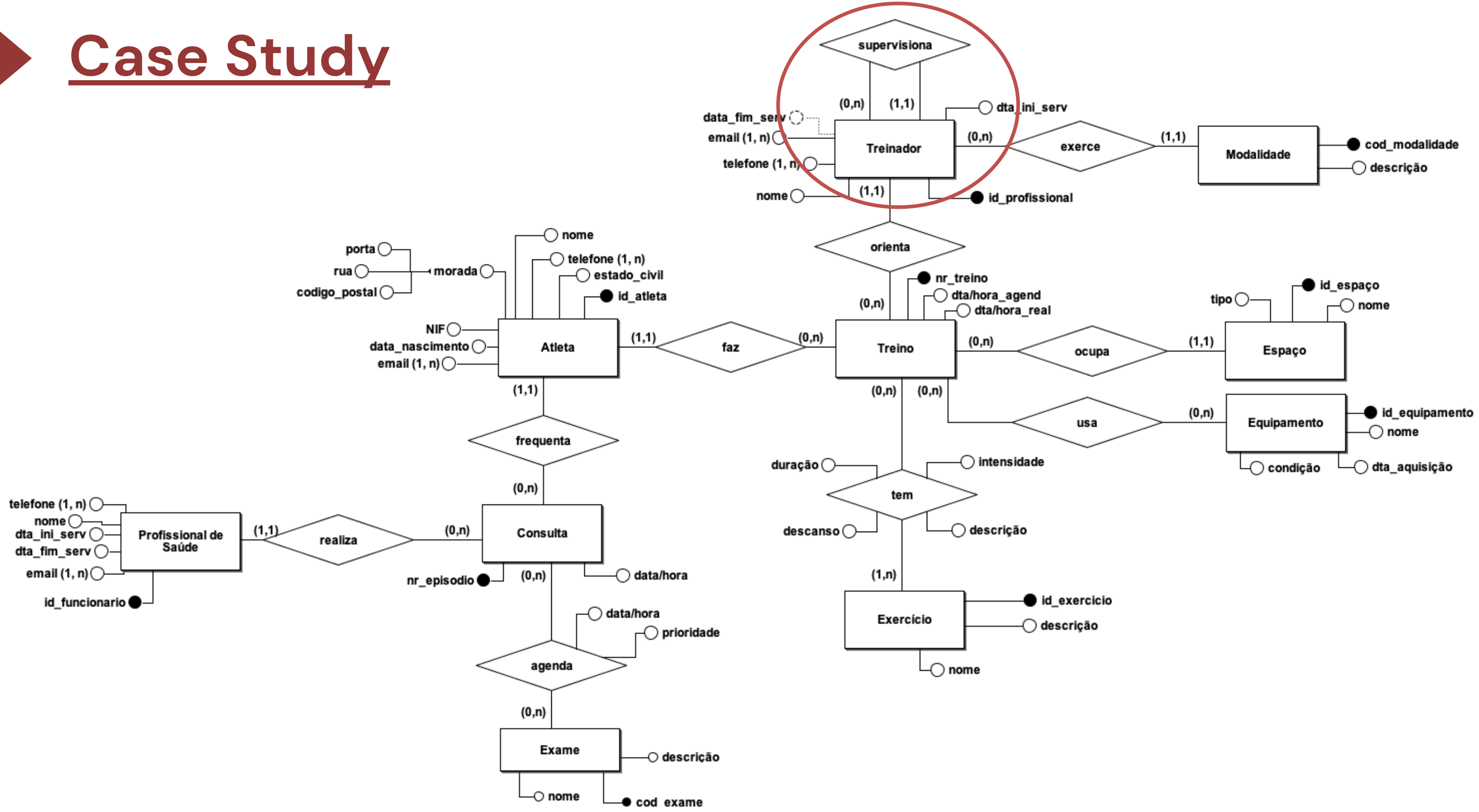
➔ Case Study

Question 10: Modify the previous E-R Diagram to make it capable of meeting the following requirement:

- A coach may be supervised by another coach.

PHASE 3: Conceptual Modeling

➜ Case Study



PHASE 3: Conceptual Modeling

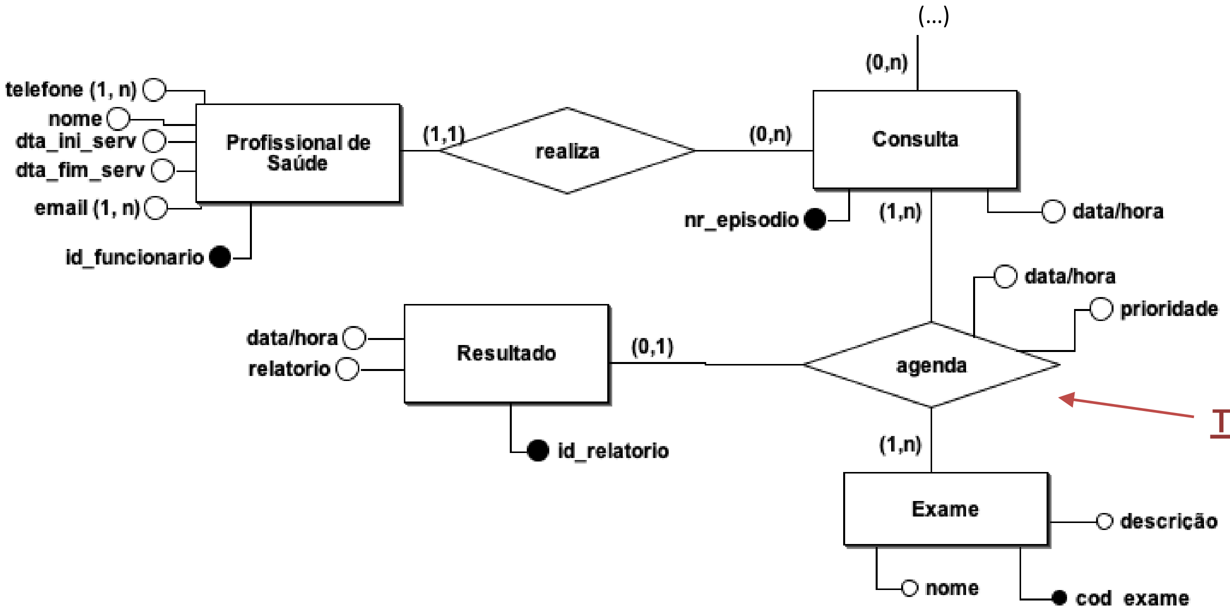
Case Study

Question 11: Modify the previous E-R Diagram to accommodate the following requirements:

- An exam scheduling may have an associated result.
- The result report can be related to different exams.
- The result must have a unique identifier (id_relatorio), a publication date, version, and a text report.

PHASE 3: Conceptual Modeling

Case Study



Ternary relationship

PHASE 3: Conceptual Modeling

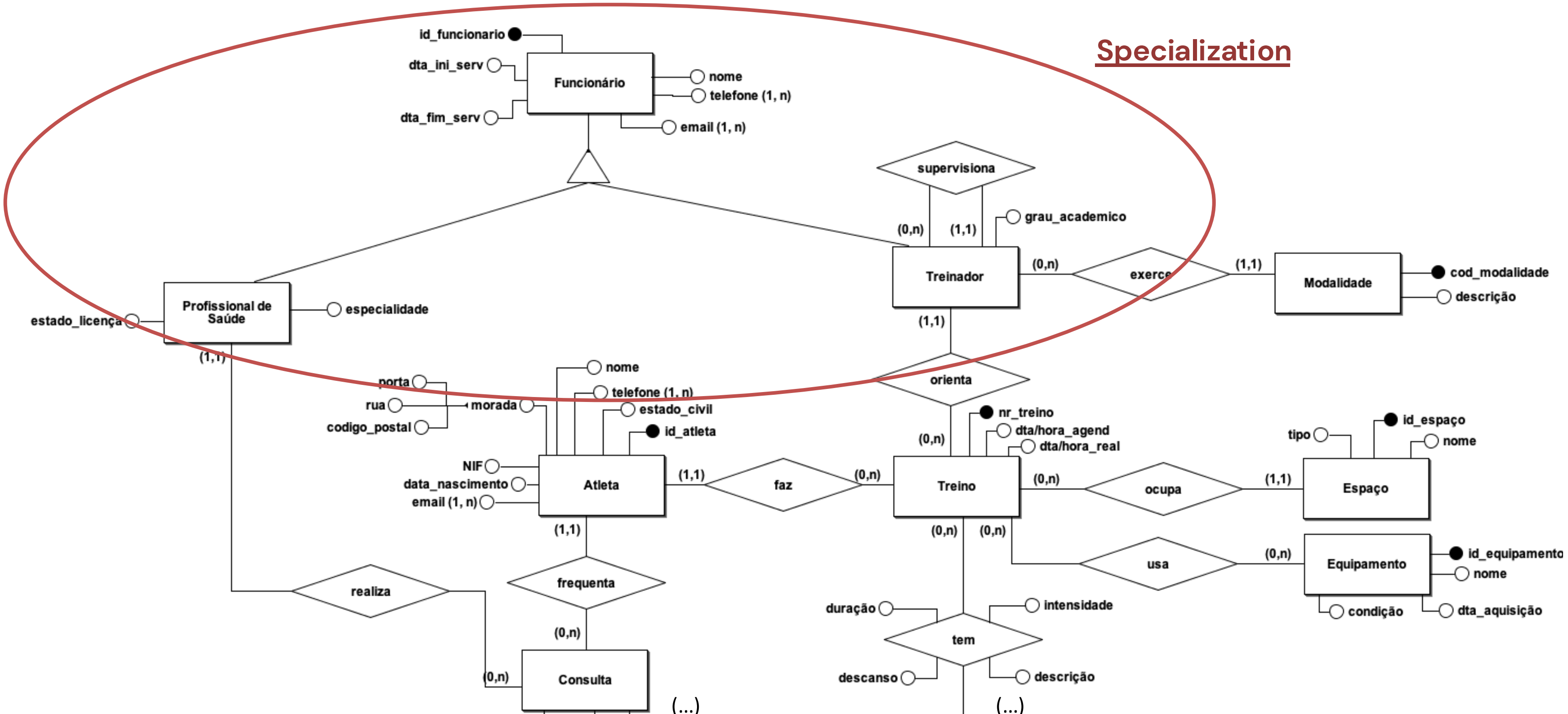
Case Study

Question 12: Build a new version of the E-R Diagram from the previous question, making use of of the concepts of enhaced modeling:

- Consider that employees can be classified as health professionals or coaches.
- All training center employees must be registered with their employee ID (unique value), name, email, phone, service start date and end date.
- Only health professionals are associated with a specialty.
- Healthcare professionals must be registered with the status of their license (active, suspended, or expired).
- Coaches are registered with the academic degree (secondary education, bachelor's, specialization, master's, doctorate).
- Only coaches are associated with one sport.

PHASE 3: Conceptual Modeling

➡ Case Study



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